Closed-loop object search | same initial scene, independent policies

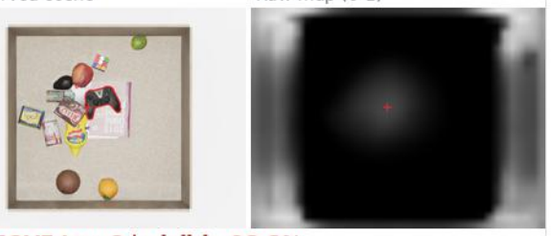
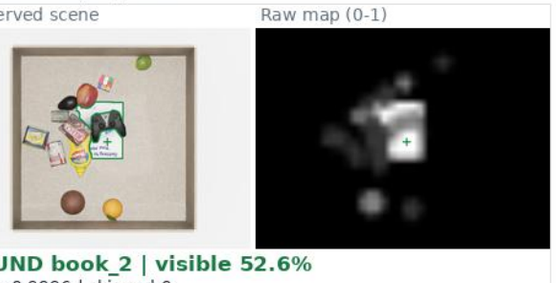
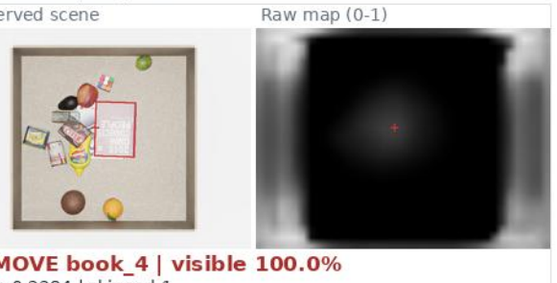
Target: book_2 Camera: center Scene: book_2_scene00009_env0026

Red: remove blocker. Green: target selected. A row is one policy decision before removal.

Object peak score -> first candidate with visibility >= 50%; step 0 is the common starting image.

Target reference



	Fusion | removals: 6	Similarity | removals: 3	Occlusion | removals: 6	Complexity | removals: 10
r=0	Observed scene / Raw map (0-1)  **REMOVE book_3 | visible 100.0%** score 0.9999 | skipped 1	Observed scene / Raw map (0-1)  **REMOVE book_3 | visible 100.0%** score 0.8547 | skipped 2	Observed scene / Raw map (0-1)  **REMOVE book_3 | visible 100.0%** score 0.6489 | skipped 0	Observed scene / Raw map (0-6)  **REMOVE book_3 | visible 100.0%** score 4.525 | skipped 2
r=1	Observed scene / Raw map (0-1)  **REMOVE toy_3 | visible 97.9%** score 1 | skipped 1	Observed scene / Raw map (0-1)  **REMOVE book_1 | visible 86.2%** score 0.8166 | skipped 2	Observed scene / Raw map (0-1)  **REMOVE book_1 | visible 86.2%** score 0.5272 | skipped 0	Observed scene / Raw map (0-6)  **REMOVE toy_1 | visible 76.2%** score 4.063 | skipped 1
r=2	Observed scene / Raw map (0-1)  **REMOVE packaged_food_4 | visible 100.0%** score 0.9998 | skipped 1	Observed scene / Raw map (0-1)  **REMOVE book_4 | visible 75.3%** score 0.8264 | skipped 1	Observed scene / Raw map (0-1)  **REMOVE toy_3 | visible 98.8%** score 0.2792 | skipped 0	Observed scene / Raw map (0-6)  **REMOVE fruit_1 | visible 100.0%** score 3.627 | skipped 1
r=3	Observed scene / Raw map (0-1)  **REMOVE packaged_food_3 | visible 100.0%** score 0.9999 | skipped 1	Observed scene / Raw map (0-1)  **FOUND book_2 | visible 52.6%** score 0.9996 | skipped 0	Observed scene / Raw map (0-1)  **REMOVE book_4 | visible 100.0%** score 0.2384 | skipped 1	Observed scene / Raw map (0-6)  **REMOVE toy_3 | visible 100.0%** score 3.417 | skipped 1
r=4	Observed scene / Raw map (0-1)  **REMOVE book_1 | visible 100.0%** score 0.9997 | skipped 2	Finished at r=3 No further observation or action	Observed scene / Raw map (0-1)  **REMOVE packaged_food_3 | visible 68.5%** score 0.06906 | skipped 0	Observed scene / Raw map (0-6)  **REMOVE packaged_food_3 | visible 92.5%** score 2.866 | skipped 1
r=5	Observed scene / Raw map (0-1)  **REMOVE book_4 | visible 100.0%** score 0.9986 | skipped 1	Finished at r=3 No further observation or action	Observed scene / Raw map (0-1)  **REMOVE fruit_1 | visible 100.0%** score 0.04872 | skipped 0	Observed scene / Raw map (0-6)  **REMOVE packaged_food_4 | visible 100.0%** score 2.106 | skipped 2
r=6	Observed scene / Raw map (0-1)  **FOUND book_2 | visible 95.5%** score 1 | skipped 0	Finished at r=3 No further observation or action	Observed scene / Raw map (0-1)  **FOUND book_2 | visible 88.7%** score 0.0484 | skipped 0	Observed scene / Raw map (0-6)  **REMOVE book_1 | visible 100.0%** score 1.811 | skipped 1
r=7	Finished at r=6 No further observation or action	Finished at r=3 No further observation or action	Finished at r=6 No further observation or action	Observed scene / Raw map (0-6)  **REMOVE fruit_2 | visible 100.0%** score 1.56 | skipped 1
r=8	Finished at r=6 No further observation or action	Finished at r=3 No further observation or action	Finished at r=6 No further observation or action	Observed scene / Raw map (0-6)  **REMOVE book_4 | visible 100.0%** score 1.376 | skipped 0
r=9	Finished at r=6 No further observation or action	Finished at r=3 No further observation or action	Finished at r=6 No further observation or action	Observed scene / Raw map (0-6)  **REMOVE packaged_food_1 | visible 100.0%** score 0.8294 | skipped 0
r=10	Finished at r=6 No further observation or action	Finished at r=3 No further observation or action	Finished at r=6 No further observation or action	Observed scene / Raw map (0-6)  **FOUND book_2 | visible 100.0%** score 0.6438 | skipped 0

After step 0, removal choices and observed scenes can differ across policies. Blank cells mean the policy has finished.

Grayscale: fixed F/S/O 0-1, C 0-6 display cap; no per-image normalization. Contours/marks use simulator IDs for action annotation.

Simulator visibility eligibility and idealized object removal; target retrieval adds one separate action. Post-hoc extreme examples.